

Maths Thermo

class # 06

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## Legendre Transforms and Helmholtz Free Energy

Defn (6.1): Suppose that  $f: [a, b] \rightarrow \mathbb{R}$  is continuous. For every  $p \in [c, d]$ , define

$$(6.1) \quad f^*(p) := \sup_{x \in [a, b]} \{xp - f(x)\}.$$

The function  $f^*: [c, d] \rightarrow \mathbb{R}$  is called the Legendre transform of  $f$ .

Theorem (6.2): Suppose that  $f \in C^2([0, \infty); \mathbb{R})$  with

$$f''(x) > 0, \quad \forall x \in [0, \infty).$$

Then, for all  $p \in \text{Range}(f')$ ,

$$(6.2) \quad f^*(p) = x_p \cdot p - f(x_p),$$

where  $x_p \in [0, \infty)$  is the unique solution to

$$f'(x_p) = p.$$

Proof: let  $p \in \text{Range}(f')$ .  $f'([0, \infty)) \xrightarrow{\text{onto}} \text{Range}(f')$ .

For every  $p \in \text{Range}(f')$ , there is a unique  $x_p \in [0, \infty)$  such that,

$$f'(x_p) = p,$$

Since  $f'$  is a strictly increasing function.

Fix  $p \in \text{Range}(f')$ . By Taylor's Theorem,

$$f(x) = f(x_p) + p(x - x_p) + \frac{1}{2}f''(\xi_p)(x - x_p)^2$$

for some  $\xi_p$  between  $x$  and  $x_p$ .

$$\sup_{0 \leq x < \infty} \{x \cdot p - f(x)\}$$

only  $x$  dependence

$$= \sup_{0 \leq x < \infty} \{x_p \cdot p - f(x_p) - \frac{1}{2}f''(\xi_p)(x - x_p)^2\}$$

$$= x_p \cdot p - f(x_p). \quad \text{///}$$

Theorem (6.3): Suppose that  $f \in C^2([0, \infty); \mathbb{R})$   
with

$$f''(x) > 0, \quad \forall x \in [0, \infty).$$

(6.3) Then,  $[f^*]'(p) = [f']^{-1}(p)$ ,  $[f^*]''(p) > 0$ ,

for all  $p \in \text{Range}(f')$ .

Proof: Recall

$$f': [0, \infty) \xrightarrow[\text{onto}]{1-1} \text{Range}(f')$$

Thus

$$f'([f']^{-1}(p)) = p, \quad \forall p \in \text{Range}(f').$$

Also, observe that  $\frac{d}{dp}([f']^{-1}) \in C(\text{Range}(f'); \mathbb{R})$ .

Now,

$$\begin{aligned}
 \frac{d}{dp} [f^*(p)] &= \frac{d}{dp} \{x_p \cdot p - f(x_p)\} \\
 &= \frac{d}{dp} \{[f']^{-1}(p) \cdot p - f([f']^{-1}(p))\} \\
 &= p \cdot \frac{d}{dp} [[f']^{-1}](p) \\
 &\quad + [f']^{-1}(p) \\
 &\quad - \underbrace{\frac{df}{dx}([f']^{-1}(p))}_{=p} \cdot \frac{d}{dp} [[f']^{-1}](p) \\
 &= [f']^{-1}(p).
 \end{aligned}$$

Now that this is established, we have

$$\frac{d^2}{dp^2} [f^*(p)] = \frac{d}{dp} [[f']^{-1}](p)$$

for all  $p \in \text{Range}(f')$ . Since  $f'' \in C([0, \infty))$  and

$$f''(x) > 0, \quad \forall x \in [0, \infty),$$

$$\frac{d}{dp} [[f']^{-1}](p) > 0, \quad \forall p \in \text{Range}(f'). \quad \text{///}$$

Example (6.4): Suppose that

$$f(x) = \exp(x), \quad x \in (-\infty, \infty).$$

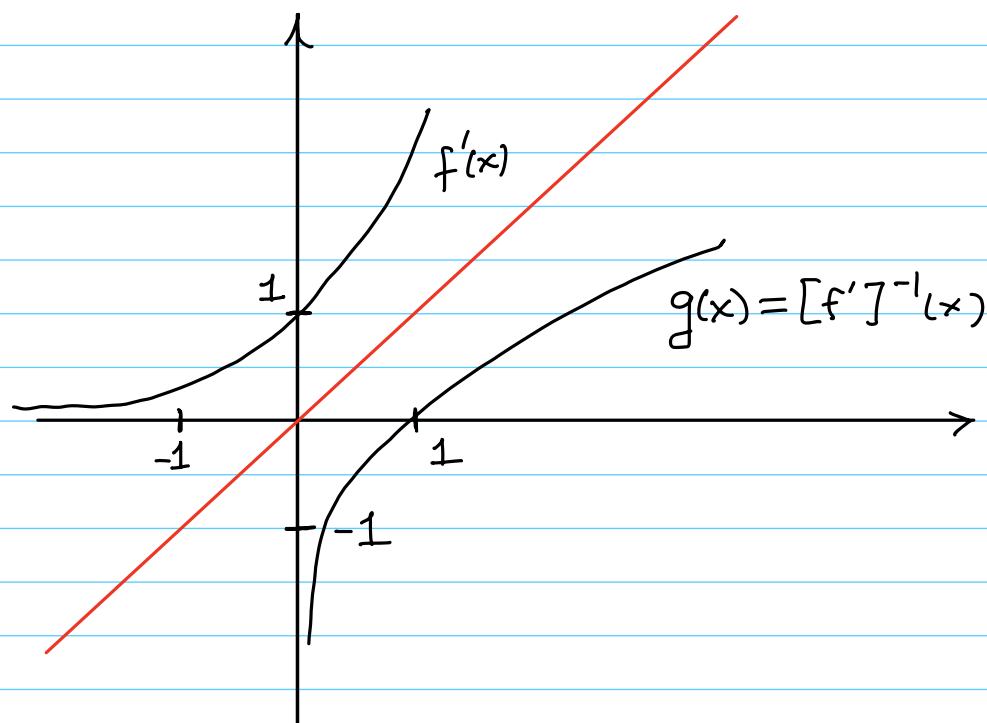
We could restrict the domain to  $[0, \infty)$ , if desired.

Then,  $f'(x) = \exp(x)$ ,  $f''(x) = \exp(x)$ .

Clearly,  $f'' > 0$  and  $f$  is convex. let us examine the derivative function,  $f'$ , because this is important

$$\text{Dom}(f') = (-\infty, \infty)$$

$$\text{Range}(f') = (0, \infty)$$



Clearly  $f': (-\infty, \infty) \xrightarrow[\text{onto}]{1-1} (0, \infty)$  is an increasing function.

$$[f']^{-1}: (0, \infty) \xrightarrow[\text{onto}]{1-1} (-\infty, \infty)$$

is

$$[f']^{-1}(p) = g(p) = \log(p).$$

Observe that

$$\begin{aligned} f'(g(p)) &= \exp(\log(p)) \\ &= p. \end{aligned}$$

For all  $p \in \text{Range}(f')$ ,

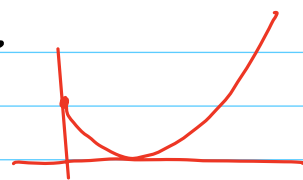
$$f^*(p) = x_p \cdot p - f(x_p), \quad x_p = [f']^{-1}(p)$$

where  $x_p \in [0, \infty)$  is the unique solution to

$$f'(x_p) = p.$$

Thus,

$$\begin{aligned} f^*(p) &= g(p)p - f(g(p)) \\ &= \log(p)p - \exp(\log(p)). \\ &= p \log(p) - p, \end{aligned}$$



for all  $p \in \text{range}(f') = (0, \infty)$ . This function

$$p \mapsto p \log(p) - p$$

is very important in Thermodynamics and is called the ideal gas function. More on this later.

We leave it to the reader to confirm that

$$[f^*]^*(x) = f(x) = \exp(x), \quad x \in \mathbb{R}.$$

We will see that this is true in general.

Theorem (6.5): Suppose that  $f \in C^2([0, \infty); \mathbb{R})$   
with

$$f''(x) > 0, \quad \forall x \in [0, \infty).$$

Then  $f^* \in C^2(\text{Range}(f'); \mathbb{R})$  with

$$[f^*]''(p) > 0, \quad \forall p \in \text{Range}(f').$$

Furthermore

(6.4) 
$$[f^*]^*(x) = f(x), \quad \forall x \in [0, \infty).$$

In other words, the Legendre transform is involutory  
and information preserving.

Proof: It follows that, for all  $x \in [0, \infty)$ ,

$$[f^*]^*(x) = x p_x - f^*(p_x),$$

where  $p_x \in \text{Range}(f')$  is the unique solution to

$$f^{*'}(p_x) = x.$$

Recall

$$f^*(p_x) = p_x \cdot x_{p_x} - f(x_{p_x}),$$

where  $x_{p_x} \in [0, \infty)$  is the unique solution to

$$f'(x_{p_x}) = p_x.$$

Of course, by uniqueness,

$$x_{p_x} = x.$$

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$$\begin{aligned} [f^*]^*(x) &= x p_x - f^*(p_x) \\ &= x \cdot p_x - \{ p_x \cdot x p_x - f(x p_x) \} \\ &= f(x). \quad /// \end{aligned}$$

Why is the Legendre transform useful in Thermodynamics?

This transform allows us to introduce new thermodynamic coordinates/variables.

Recall

$$\tilde{U} = \tilde{U}(S, V, N)$$

or

$$d\tilde{U} = T dS - P dV + \mu dN$$

The latter means, in short hand,

$$T = \frac{\partial \tilde{U}}{\partial S}, \quad -P = \frac{\partial \tilde{U}}{\partial V}, \quad \mu = \frac{\partial \tilde{U}}{\partial N}.$$

$T, -P, \mu$  are called conjugate variables.

| Extensive Quantity | Conjugate Variable | Variable |
|--------------------|--------------------|----------|
| $\tilde{U}$        | $T$                | $S$      |
|                    | $-P$               | $V$      |
|                    | $\mu$              | $N$      |
| $\tilde{S}$        | $1/T$              | $U$      |
|                    | $P/T$              | $V$      |
|                    | $-\mu/T$           | $N$      |

Definition (6.6): Suppose that all of the Thermodynamic Postulates hold. Assume that the material in question is unary ( $r=1$ ). Define

$$(6.5) \quad \tilde{F}(T, V, N) = \tilde{U}(S_T, V, N) - TS_T$$

where  $S_T = S_T(V, N)$  is the unique solution of the equation

$$(6.6) \quad T_U(S_T(V, N), V, N) = T$$

for fixed values of  $N$  and  $V$ .

$\tilde{F}$  is called the Helmholtz free energy.

Formally, we find, taking the differential of

$$\tilde{F} = U - TS,$$

that

$$d\tilde{F} = d\tilde{U} - SdT - TdS$$

$$= TdS - PdV + \mu dN - SdT - TdS$$

$$= -SdT - PdV + \mu dN.$$

| Extensive Quantity | Conjugate Variables   | Variable          |
|--------------------|-----------------------|-------------------|
| $\tilde{U}$        | $T$<br>$-P$<br>$\mu$  | $S$<br>$V$<br>$N$ |
| $\tilde{F}$        | $-S$<br>$-P$<br>$\mu$ | $T$<br>$V$<br>$N$ |



Example (6.7): Suppose that

$$\tilde{S}(U, V, N) = \left( \frac{N V U R^2}{v_0 \theta} \right)^{1/3}.$$

It follows that

$$\tilde{U}(S, V, N) = \frac{S^3 v_0 \theta}{N V R^2}.$$

The temperature function is

$$T_U(S, V, N) = \frac{3 S^2 v_0 \theta}{N V R^2}.$$

Or

$$\begin{aligned} \frac{1}{T_S(U, V, N)} &= \frac{1}{3} \left( \frac{N V U R^2}{v_0 \theta} \right)^{-2/3} \frac{N V R^2}{v_0 \theta} \\ &= \frac{1}{3} \left( \frac{N V R^2}{v_0 \theta} \right)^{1/3} U^{-2/3}. \end{aligned}$$

Thus

$$T_S(U, V, N) = 3 \left( \frac{v_0 \theta}{N V R^2} \right)^{1/3} U^{2/3}.$$

Recall that

$$T_U(S, V, \vec{N}) = T_S(\tilde{U}(S, V, N), V, \vec{N})$$

and

$$T_S(U, V, \vec{N}) = T_U(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

By definition

$$\tilde{F}(T, V, N) = \tilde{U}(S_T, V, N) - T S_T$$

where

$$T_U(S_T(V, N), V, N) = T.$$

For our example,

$$T_U(S, V, N) = 3 \frac{S^2 v_0 \theta}{N V R^2} = T.$$

Hence,

$$S_T(V, N) = \sqrt{\frac{T N V R^2}{3 v_0 \theta}}.$$

Thus,

$$\begin{aligned} \tilde{F}(T, V, N) &= \tilde{U} \left( \sqrt{\frac{T N V R^2}{3 v_0 \theta}}, V, N \right) \\ &\quad - T \sqrt{\frac{T N V R^2}{3 v_0 \theta}} \end{aligned}$$

But

$$\begin{aligned} \tilde{U}(S_T, V, N) &= \frac{S_T^3 v_0 \theta}{N V R^2} \\ &= \left( \frac{T N V R^2}{3 v_0 \theta} \right)^{3/2} \frac{v_0 \theta}{N V R^2} \end{aligned}$$

So that

$$\begin{aligned} \tilde{F}(T, N, V) &= \left( \frac{T N V R^2}{3 v_0 \theta} \right)^{3/2} \frac{v_0 \theta}{N V R^2} \\ &\quad - T \sqrt{\frac{T N V R^2}{3 v_0 \theta}} \end{aligned}$$

$$= \frac{1}{3} \frac{T^{3/2}}{\sqrt{3}} \sqrt{\frac{NVR^2}{v_0 \theta}} - \frac{T^{3/2}}{\sqrt{3}} \sqrt{\frac{NVR^2}{v_0 \theta}}$$

Finally,

$$\tilde{F}(T, V, N) = -\frac{2T^{3/2}}{3} \sqrt{\frac{NVR^2}{3v_0 \theta}}.$$

Now,

$$\frac{\partial \tilde{F}}{\partial T} = -\frac{2}{3} \cdot \frac{3}{2} \cdot \sqrt{T} \cdot \sqrt{\frac{NVR^2}{3v_0 \theta}}.$$

But

$$S_T(V, N) = \sqrt{\frac{TNVR^2}{3v_0 \theta}}.$$

and

$$\frac{\partial \tilde{F}}{\partial T} = -S_T.$$

Theorem (6.8): Suppose that all of the Thermodynamic Postulates hold. Assume that the material in question is unary ( $r=1$ ). The following provides an equivalent means for finding the Helmholtz free energy:

$$\tilde{F}(T, V, N) = U_T - T \tilde{S}(U_T, V, N),$$

where  $U_T = U_T(V, N)$  is the unique solution of the equation

$$T_S(U_T(V, N), V, N) = T,$$

for fixed values of  $N$  and  $V$ .

Proof: Exercise III

Example (6.9): let us use the second version to calculate the free energy for the monatomic ideal gas. Recall,

$$\tilde{S}(U, V, N) = N s_0 + NR \ln \left( \left( \frac{U}{U_0} \right)^{3/2} \left( \frac{V}{V_0} \right) \left( \frac{N}{N_0} \right)^{-5/2} \right),$$

where  $s_0, R, U_0, V_0, N_0$  are positive constants.

Then

$$\frac{\partial \tilde{S}}{\partial U} = \frac{NR}{\left( \frac{U}{U_0} \right)^{3/2}} \cdot \frac{3}{2} \left( \frac{U}{U_0} \right)^{1/2} \frac{1}{U_0}$$

$$= \frac{3NR}{2U_0} \cdot \frac{U_0}{U}$$

$$= \frac{3NR}{2U}$$

Thus

$$T_S(U, V, N) = \frac{2}{3} \frac{U}{NR}$$

and

$$U_T(N, V) = \frac{3NRT}{2}.$$

Therefore,

$$\begin{aligned}\tilde{F}(T, V, N) &= \frac{3NRT}{2} - TNs_0 \\ &\quad - NRT \ln \left( \left( \frac{3NRT}{2U_0} \right)^{3/2} \left( \frac{V}{V_0} \right) \left( \frac{N}{N_0} \right)^{-5/2} \right)\end{aligned}$$

Define

$$T_0 := \frac{2U_0}{3N_0R}.$$

Then,

$$\begin{aligned}\tilde{F}(T, V, N) &= \frac{3NRT}{2} - TNs_0 \\ &\quad - NRT \ln \left( \left( \frac{T}{T_0} \right)^{3/2} \left( \frac{V}{V_0} \right) \left( \frac{N}{N_0} \right)^{-1} \right).\end{aligned}$$

What are the derivatives  $\frac{\partial \tilde{F}}{\partial T}$ ,  $\frac{\partial \tilde{F}}{\partial V}$ ,  $\frac{\partial \tilde{F}}{\partial N}$ ?

As an exercise, show that

$$\frac{\partial \tilde{F}}{\partial T} = -\tilde{S}(U_T(V, N), V, N) = -S_T(V, N),$$

$$\frac{\partial \tilde{F}}{\partial V} = -P_s(U_T(V, N), V, N) = -P_v(S_T(V, N), V, N),$$

$$\frac{\partial \tilde{F}}{\partial N} = \mu_s(U_T(V, N), V, N) = \mu_v(S_T(V, N), V, N).$$

Are these relations true in general?